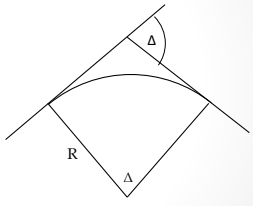
INS3441
HIGHWAY ENGINEERING

Assoc. Prof. Halit ÖZEN, PhD.

WEEK	SUBJECT	DATE	
1	INTRODUCTION BASIC DEFINITIONS (TERMS)	05.10.2020 08.10.2020	Monday Thursday
2	ROAD USER CHARECTERISTICS VEHICLE MOVEMENTS	12.10.2020 15.10.2020	Monday Thursday
3	TRAFFIC FLOW AND CAPACITY ROADWAY CLASSIFICATION AND GEOMETRIC STANDARDS	19.10.2020 22.10.2020	Monday Thursday
4	HIGHWAY ROUTE STUDY **NATIONAL HOLIDAY**	26.10.2020 29.10.2020	Monday Thursday
5	HORIZANTAL CURVE AND SUPERELEVATION (Project sheet distribution) HORIZANTAL CURVE AND SUPERELEVATION	02.11.2020 05.11.2020	Monday Thursday
6	APPLICATION (VEHICLE MOVEMENTS) ASSIGNMENT - APPLICATION (TRAFFIC)	09.11.2020 12.11.2020	Monday Thursday
7	APPLICATION (HORIZANTAL CURVE) ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)	16.11.2020 19.11.2020	Monday Thursday
8	**MID TERM EXAM**	23.11.2020 26.11.2020	Monday Thursday
9	TRANSITION CURVES- ASSIGNMENT (HAND IN) TRANSITION CURVES – ASSIGNMENT (RETURN)	30.11.2020 03.12.2020	Monday Thursday
10	VERTICAL ALIGNMENT AND VERTICAL CURVE VERTICAL ALIGNMENT AND VERTICAL CURVE	07.12.2020 10.12.2020	Monday Thursday
11	APPLICATION (TRANSITION CURVES) ASSIGNMENT	14.12.2020 17.12.2020	Monday Thursday
12	APPLICATION (VERTICAL CURVE) APPLICATION (VERTICAL AND HORIZANTAL CURVE)	21.12.2020 24.12.2020	Monday Thursday
13	PLANNING OF INTERSECTION AND URBAN ROADS PROJECTS DELIVERY DUE DATE (December 30, 2020)	28.12.2020 31.12.2020	Monday Thursday
14	DRAINAGE IN HIGHWAY **MAKEUP EXAM**	04.01.2020 07.01.2020	Monday Thursday

HORIZANTAL CURVES

- Δ– Intersection angle
- R - Radius



HORIZANTAL CURVES

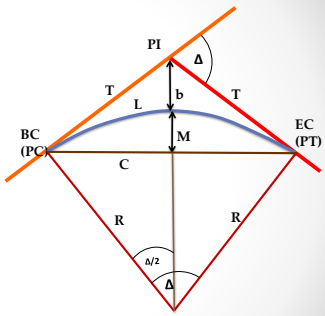
Type of the Horizantal Curves

- **Simple**
- **Compound**
- **Inverse**

HORIZANTAL CURVES

Simple Curve

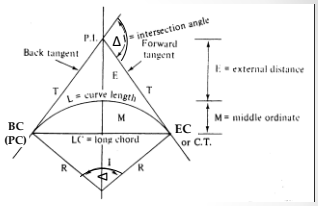
Δ = Deflection Angle
R = Radius of Circular Curve
D = Length of Curvature
(D = EC – BC)
BC = Beginning of Curve (PC)
EC = End of Curve (PT)
PI = Point of Intersection
t = Tangent Length
(t = PI – BC = EC - PI)
C = Chord Length
M = Middle Ordinate
b = External Distance



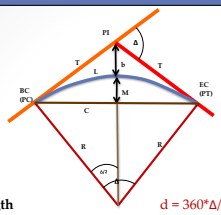
HORIZANTAL CURVES

- BC/PC: Point of Curvature
- BC = PI – T
  - PI = Point of Intersection
  - T = Tangent

- EC/PT: Point of Tangency
- EC = BC + L
  L = Length



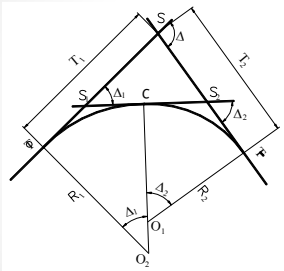
HORIZANTAL CURVES



- **Length** d = 360*Δ/D **(3)**
- **Tangent** T = R*Tan(Δ /2) **(4)**
- **Chord** C = 2R*Sin(Δ /2) **(5)**
- **External Distance** b = R*(Sec(Δ/2) – 1) **(6)**
- **Mid Ordinate** M = (R-R*Cos(Δ /2)) **(7)**

HORIZANTAL CURVES

Compound Curve



R_1, Δ_1, b_1, t_1

R_2, Δ_2, b_2, t_2

$\boldsymbol{\Delta = \Delta_1 + \Delta_2}$

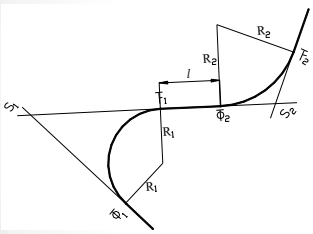
$\overline{S_1S_2} = t_1 + t_2$

$S_2 = t_2 = R_2 * \tan\left(\frac{\Delta_2}{2}\right)$

$$\frac{SS_1}{Sin\Delta_2} = \frac{SS_2}{Sin\Delta_1} = \frac{S_1S_2}{Sin\Delta}$$

HORIZANTAL CURVES

Inverse Curve



$\overline{S_1S_2} = t_1 + l + t_2$

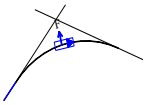
$t_1 = R_1 * \tan\frac{\Delta_1}{2}$

$t_2 = R_2 * \tan\frac{\Delta_2}{2}$

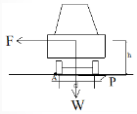
$l \geq 60m$

HORIZANTAL CURVES

Vehicle stability on horizontal curves



F: centrifugal force



W: Vehicle weigth
R: Curve Radii
μ_e: Lateral frictional coefficient
P: Lateral frictional force

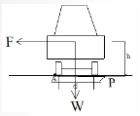
Lateral sliding

$$F \le P \qquad P = N * \mu_e \qquad N = W \qquad P = W * \mu_e$$

$$m * \frac{V^2}{R} = m * g * \mu_e \qquad V_{sliding} = \sqrt{\mu_e * R * g}$$

HORIZANTAL CURVES

Vehicle stability on horizontal curves



W: Vehicle weigth
R: Curve Radii
μ_e: Lateral frictional coefficient
P: Lateral frictional force

Overturning

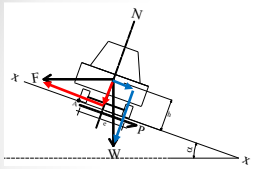
$$\sum M_A = 0 \qquad F * h = W * \frac{d}{2} \qquad m\frac{V^2}{R} * h = m * g * \frac{d}{2} \qquad V_{overturning} = \sqrt{\frac{d * R * g}{2 * h}}$$

$$V_{overturning}(km/sa) = 7{,}97 * \sqrt{\frac{R * d}{h}}$$

$$V_{sliding} = V_{overturning} \qquad \sqrt{\mu_e * R * g} = \sqrt{\frac{d * R * g}{2 * h}} \qquad \mu_e = \frac{d}{2 * h}$$

HORIZANTAL CURVES

Vehicle on Curved Banked Track



Lateral sliding

$$\sum_{x-x} = 0 \qquad P = \mu_e * N$$

$$N = W * Cos\alpha + F * Sin\alpha = W * Cos\alpha + \frac{W}{g} * \frac{V^2}{R} * Sin\alpha$$

$$F * Cos\alpha = W * Sin\alpha + P$$

$$P = \mu_e * N = \mu_e * (W * Cos\alpha + \frac{W}{g} * \frac{V^2}{R} * Sin\alpha)$$

$$\frac{V^2}{g * R} - \frac{V^2}{g * R} * tasn\alpha * \mu_e = \tan\alpha + \mu_e$$

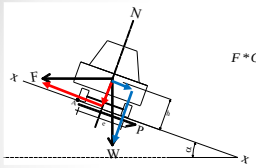
$$V_{sliding} = \sqrt{\frac{g * R * (\tan\alpha + \mu_e)}{1 - \tan\alpha * \mu_e}}$$

$$V_{sliding}(km/sa) = 11{,}28 * \sqrt{\frac{R * (\tan\alpha + \mu_e)}{1 - \tan\alpha * \mu_e}}$$

HORIZANTAL CURVES

Vehicle on Curved Banked Track

Overturning



$$\sum M_A = 0$$

$$F*Cos\alpha*h = F*Sin\alpha*\frac{e}{2} + W*Cos\alpha*\frac{e}{2} + W*Sin\alpha*h$$

$$\frac{V^2}{g*R}*\left(h - \frac{e}{2}*\tan\alpha\right) = \frac{e}{2} + h*\tan\alpha$$

$$V_{max} = \sqrt{\frac{h*\tan\alpha + \frac{e}{2}}{h - \tan\alpha*\frac{e}{2}}*R*g}$$

$$V_{max}(km/sa) = 11{,}28\sqrt{\frac{R*\left(h*\tan\alpha + \frac{e}{2}\right)}{h - \tan\alpha*\frac{e}{2}}}$$

HORIZANTAL CURVES

Minumum Radii for Curved Banked Track

$$V_{dizayn}(km/sa) = 11{,}28*\sqrt{\frac{R*(\tan\alpha + \mu_e)}{1 - \tan\alpha*\mu_e}}$$

$$R_{min(dizayn)}(m) = \frac{V^2*(1 - \mu_e*\tan\alpha)}{127{,}14*(\mu_e + \tan\alpha)}$$

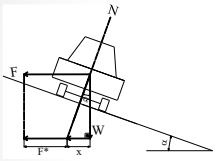
$$V_{güvenli}(km/sa) = 11{,}28\sqrt{\frac{R*\left(h*\tan\alpha + \frac{e}{2}\right)}{h - \tan\alpha*\frac{e}{2}}}$$

$$R_{min(güvenli)}(m) = \frac{V^2*\left(h - \frac{e}{2}*\tan\alpha\right)}{127{,}14*\left(h*\tan\alpha + \frac{e}{2}\right)}$$

HORIZANTAL CURVES

Lateral acceleration and jerk

Vehicles moving on a super-elevated horizontal curve are affected by the lateral acceleration caused by the centrifugal force.



p : lateral acceleration
tanα (q) : superelevation
F* : force that causes the lateral acceleration

$$F^* + x = F$$

$$x = W*\tan\alpha$$

$$m*\frac{V^2}{R} = F^* + m*g*\tan\alpha$$

$$F^* = m*\frac{V^2}{R} - m*g*\tan\alpha$$

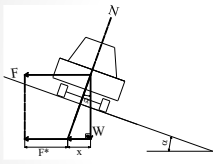
$$F^* = m*p = m*\frac{V^2}{R} - m*g*\tan\alpha$$

$$p = \frac{V^2}{R} - g*\tan\alpha$$

HORIZANTAL CURVES

Lateral acceleration and jerk

Vehicles moving on a super-elevated horizontal curve are affected by the lateral acceleration caused by the centrifugal force.



p : lateral acceleration (m/ sec^2)
tanα (q) : superelevation
F* : force that causes the lateral acceleration

$$p = \frac{V^2}{R} - g * \tan\alpha$$

$$p = \frac{V^2(km/sa)}{12{,}96 * R(m)} - 9{,}81 * \tan\alpha$$

HORIZANTAL CURVES

The lateral slope, given to the highway cross-section to counter the skid and the rollover effects of centrifugal force while travelling on a horizontal curve with a specific velocity, is named as the **superelevation**.

An important element contributing to the stability is the latidunal friction. If an ideal situation with no latidunal friction is assumed, the whole centrifugal force is to be countered by the superelevation. When μ_e=0 is written, is obtained.

$$p = \frac{V^2(km/sa)}{12{,}96 * R(m)} - 9{,}81 * q$$

If lateral friction equals to zero, that means there is not any effects (which also lateral accelaration (p) is «0») any person that inside the vehicle.

$$q_t(\%) = 0{,}786 * \frac{V^2(km/sa)}{R(m)}$$

Theoritical Superelevation

Maximum lateral acceleration **p≤1,47 m/sec^2** (German RAL Guide).
p≤0,65 m/sec^2 (Megyeri, 1993)

HORIZANTAL CURVES

Maximum Superelevation Rates

1. Twelve percent (12%) is the maximum superelevation rate in use. Drivers feel uncomfortable on sections with higher rates, and driver effort to maintain lateral position is high when speeds are reduced on such curves.
2. Where snow and ice are prevalent, a maximum value of 8% is generally used. Many agencies use this as an upper limit regardless, due to the effect of rain or mud on highways.
3. In urban areas where speeds may be reduced 3. In urban areas, where speeds may be reduced frequently due to congestion, maximum rates of 4%-6% are often used.
4. On low-speed urban streets or at intersections, superelevation may be eliminated.

HORIZANTAL CURVES

On the other hand, on real life conditions, friction may be low but its effects can not be neglected. Thus, the contribution of the lateral friction to counter the effect of the centrifugal force can not be neglected.

In some countries, half of the effect of the centrifugal force is countered by the superelevation, while the remaining half by the lateral friction. The superelevation distance calculated this way is named as the **practical superelevation**.

$$q_{pra} = 0{,}786 \cdot \frac{V_p^2}{R} \cdot 0{,}50 = 0{,}393 \cdot \frac{V_p^2}{R}$$

General Directorate of Highway (KGM)

$$q_t(\%) = 0{,}443 * \frac{V^2}{R}$$

%56 of the centrifugal force is countered by superelevation, and **%44** is countered by lateral friction

HORIZANTAL CURVES

Jerk is a value that used to determine the voyage comfort and is known as comfort criterion while designing of road.
Jerk is to change of acceleration in a unit of time.

$$p' = \frac{p}{t} \qquad V = \frac{L}{t} \qquad t = \frac{L}{V} \qquad P' = \frac{p}{\frac{L}{V}} = \frac{p * L}{V} \ (m/sec^3)$$

$$p' = \frac{V^3(km/sa)}{46{,}7 * R(m) * L(m)} - \frac{V(km/sa) * q}{36{,}7 * L(m)}$$

Most of the current agencies are used as reference of route comfort design, considering that;

- $p' \leq 0{,}4$ m/s^3 is an acceptable level,
- $0{,}4$ m/s$^3 \leq p' \leq 0{,}6$ m/s^3 is able to endure, and
- $p' \geq 0{,}6$ m/s^3 is beyond human endurance.